

Apocrine Bromhidrosis

Apocrine secretions are responsible for odor production, primarily through bacterial action on their components. The axilla hosts a diverse microbiome, predominantly composed of gram-positive bacteria. Odor formation in apocrine bromhidrosis is particularly associated with aerobic *Corynebacterium* species. Odorous steroids, including **16-androstenes**, **5 α -androstanol**, and **5 α -androstenone**, are known contributors to the characteristic malodor of bromhidrosis.[3,4] The biotransformation of these steroids involves complex metabolic pathways that are not yet fully understood and require further research.[3,4]

Bacterial degradation of apocrine secretions yields **ammonia** and **short-chain fatty acids**. The most well-characterized of these is **5 ϵ -3-methyl-2-hexenoic acid**, which is transported to the skin surface bound to two specific proteins: **apocrine secretion-binding proteins (ASOB1 and ASOB2)**. ASOB2 has been identified as **apolipoprotein D**. [5,6]

5 α -Reductase type I is expressed in apocrine glands, and individuals with bromhidrosis exhibit increased levels of this enzyme. Since 5 α -reductase catalyzes the conversion of **testosterone** to **dihydrotestosterone (DHT)**, affected individuals may have higher DHT concentrations than testosterone in the skin.

The role of **hyperhidrosis** (excessive eccrine sweating) in apocrine bromhidrosis remains unclear. Some suggest that increased eccrine sweat may reduce odor by flushing out apocrine secretions. Conversely, others propose that eccrine sweat exacerbates bromhidrosis by spreading apocrine components across the skin and maintaining a moist environment conducive to bacterial proliferation.[2]

Clinical Findings

History

Patients typically report an unpleasant body odor. The **axillae** are the most commonly affected site, though involvement of the **genital area** or **plantar surface of the feet** can also occur. Diagnosis is primarily clinical. Perception of "normal" body odor varies widely among individuals and ethnic groups. In some Asian populations, even mild odor may be considered abnormal or diagnostic.[8,9]

Cutaneous Lesions

Physical examination is usually unremarkable, with no visible skin changes.

Laboratory Tests

No laboratory abnormalities are associated with apocrine bromhidrosis.

Pathology

Findings are variable. Some studies report no structural abnormalities in apocrine glands, while others describe an **increase in the number and size** of apocrine glands in affected individuals.[8]

Differential Diagnosis

Apocrine bromhidrosis must be distinguished from **eccrine bromhidrosis**, which is less common (see Chap. 82). Eccrine sweat is generally **odorless**, widely distributed, and functions in thermoregulation. Eccrine bromhidrosis may arise from bacterial breakdown of **keratin softened by sweat**, and is typically localized to the **plantar feet**. Contributing factors include certain **foods** (e.g., garlic, curry, alcohol), **drugs** (e.g., bromides), **toxins**, or **metabolic disorders** (e.g., amino acid metabolism defects).

Treatment

General Measures

- Frequent washing of the axillae
- Use of **deodorants** or **antiperspirants** (e.g., aluminum chloride)
- Maintaining good hygiene to reduce bacterial load

(Note: Image reference removed as per content guidelines.)